GREEN TEA LCA

Forest Residue Bioenergy & SAF Techno-Economic Analysis

Scenario	Current Scenario
Mode	SAF
Generated	2026-04-28 17:55

NPV (Equity)	$-167.59M	**TCI**	$1197.78M
Equity IRR	7.13%	**SAF Output (Yr1)**	29.08M gal/yr
MFSP — SAF	$5.3267/gal	**Total Biomass**	917.4k odt/yr

1. Supply Chain & Transport

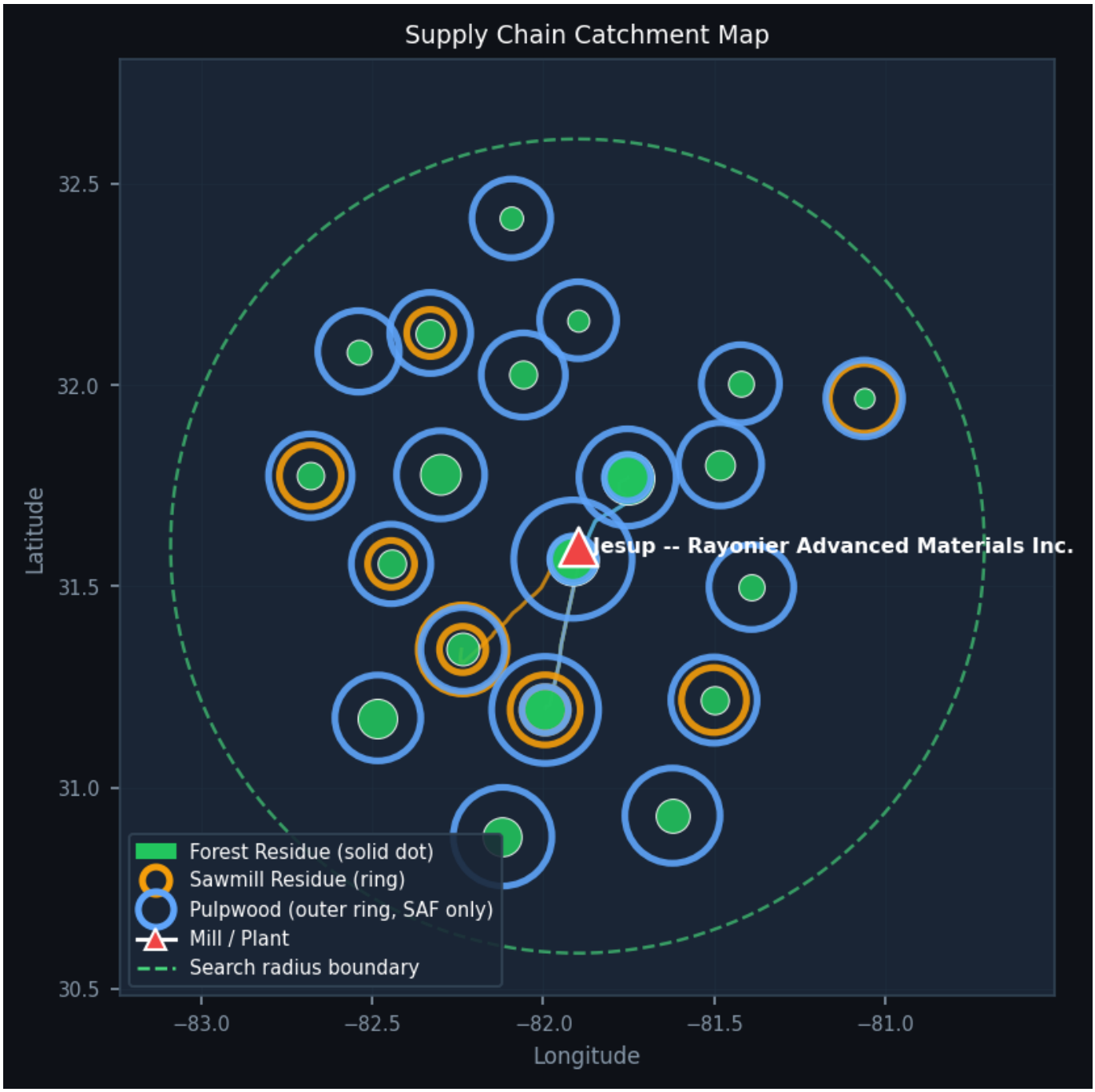
Source	Counties	Base Supply (k odt/yr)	Obtainability	Effective (k odt/yr)	Haul Dist (mi)	Transport Opt.	Delivered ($/ODT)
Forest Residue	3	143.2k	100%	99.1k	16	1.1	$63.20
Sawmill Residue	2	386.0k	100%	386.0k	32	3.1	$25.24
Pulpwood	3	432.3k	100%	432.3k	14	4.1	$27.23

Mill / Plant	Jesup -- Rayonier Advanced Materials Inc.
Search Radius	70 mi
Coordinates	31.6000°N, -81.9000°W

Transport Inputs

Parameter	Value
Avg. Truck Speed	20 mph
Forest Transport Option	-
Sawmill Transport Option	-
Pulpwood Transport Option	-

Forest Obtainability	100%
Sawmill Obtainability	100%
Pulpwood Obtainability	100%



Supply catchment map — Jesup -- Rayonier Advanced Materials Inc., 70-mile radius. Green: forest residue, amber: sawmill residue, blue: pulpwood. Dot size ∝ supply volume. Dashed circle = search boundary.

2. Economic Performance

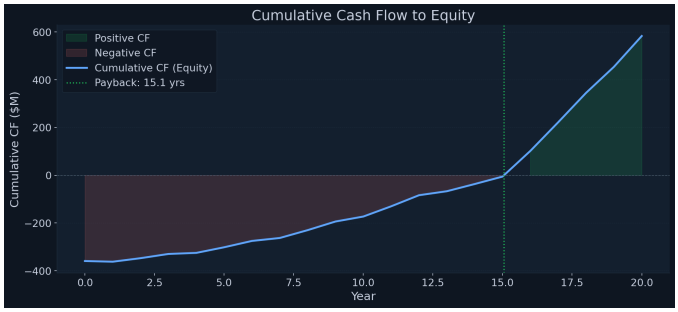
Inputs

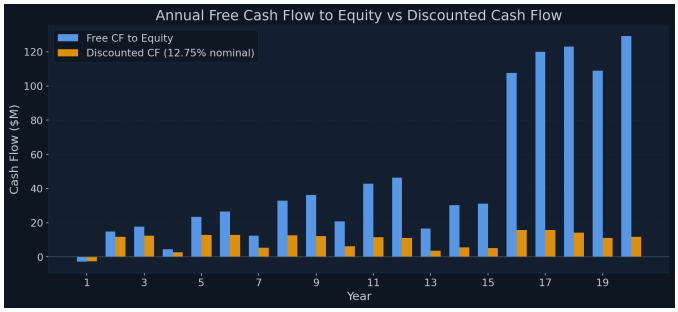
Analysis Year	2025	**Discount Rate**	10.0%
Distillate	distillate 1	**Inflation**	2.5%
SAF Price	$6.09/gal	**Debt Fraction**	70%
Diesel Price	$3.90/gal	**Loan Rate**	8.0%
Naphtha Price	$2.84/gal	**Loan Term**	15 yr
Plant Life	20 yr	**Federal Tax**	21%
CPI	321.05	**State Tax**	7%
Degradation	0.0%/yr	**Price Escalation**	2.5%/yr
		Fuel Escalation	2.5%/yr
		Cost Escalation	2.5%/yr

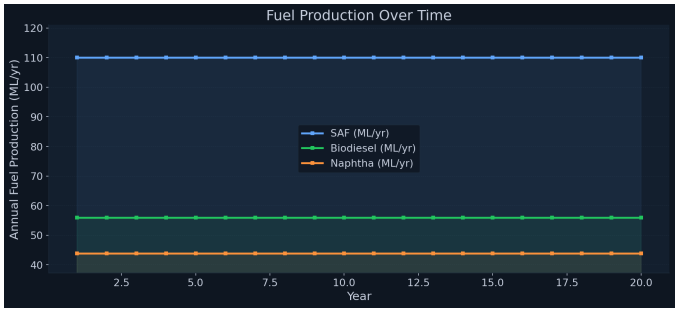
Outputs

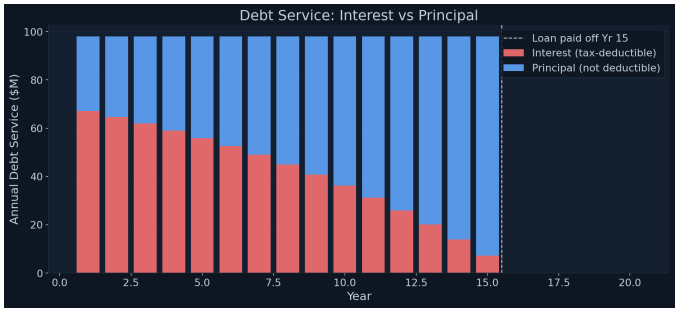
Metric	Value	Notes
TCI	$1197.78M	Total Capital Investment
FCI	$1041.55M	Fixed Capital Investment
Year 1 Revenue	$267.65M/yr	
NPV (Equity)	$-167.59M	Negative = below hurdle rate
Equity IRR	7.13%	
Payback Period	15.1 yr	
MFSP — SAF	$5.3267/gal	At NPV=0
MFSP — Diesel	$5.4143/gal	At NPV=0
MFSP — Naphtha	$4.9454/gal	At NPV=0
SAF Production	29.08M gal/yr	Year 1
Diesel Production	14.79M gal/yr	Year 1
Naphtha Production	11.58M gal/yr	Year 1

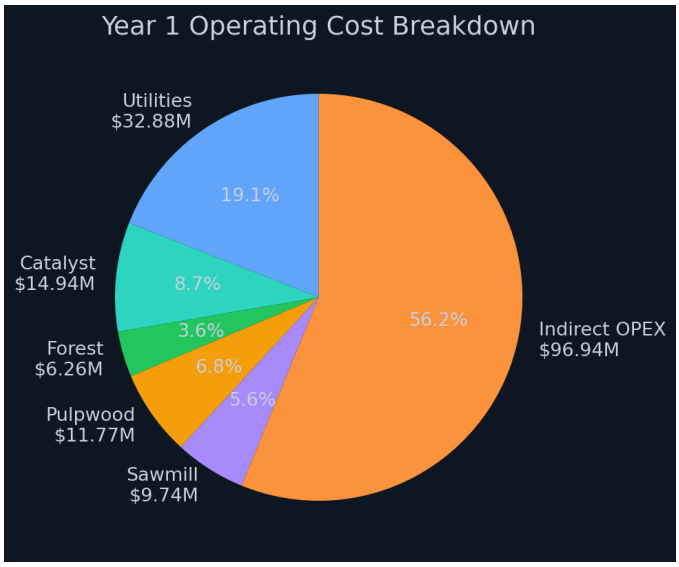
Cash Flow Plots






Fuel Production Over Time
Annual Fuel Production (ML/yr)
Year
SAF (ML/yr)
Biodiesel (ML/yr)
Naphtha (ML/yr)


Debt Service: Interest vs Principal
Annual Debt Service ($M)
Year
Loan paid off Yr 15
Interest (tax-deductible)
Principal (not deductible)


Year 1 Operating Cost Breakdown
Utilities
$32.88M
19.1%
Catalyst
$14.94M
8.7%
Forest
$6.26M
3.6%
Pulpwood
$11.77M
6.8%
Sawmill
$9.74M
5.6%
Indirect OPEX
$96.94M
56.2%

3. Lifecycle GHG Emissions

Emission Factor Inputs

Parameter	Value	Notes
Moisture Content	40%	Feedstock
Truck Fuel (L/t-mi)	0.000094	Transport EF
Transport CO_2 EF	2.680 kg/L	Diesel combustion
Transport CH_4 EF	2.7186×10^{-5} kg/L	
Transport N_2O EF	3.6488×10^{-6} kg/L	
Processing CO_2 EF	2.680 kg/L	Diesel
Processing NCV	35.8 MJ/L	Diesel
Processing CH_4 EF	0.100 g/MJ	
Processing N_2O EF	0.100 g/MJ	

Stage	Biogenic CO_2 (t/yr)	Fossil CO_2 (t/yr)	CH_4 CO_2e (t/yr)	N_2O CO_2e (t/yr)	Total (t CO_2e/yr)
Processing	0	5,654	211	2,062	7,926
Transport	0	1,351	0	1	1,352
SAF Production	969,815	466	107	1,046	971,433
TOTAL	969,815	7,471	318	3,108	980,712

Biogenic CO_2 excluded per IPCC (2006) / RED II. Section 4 uses non-biogenic totals only — consistent with the open-burning baseline EF (0.143 kg CO_2e/kg OD) which likewise excludes combustion CO_2.

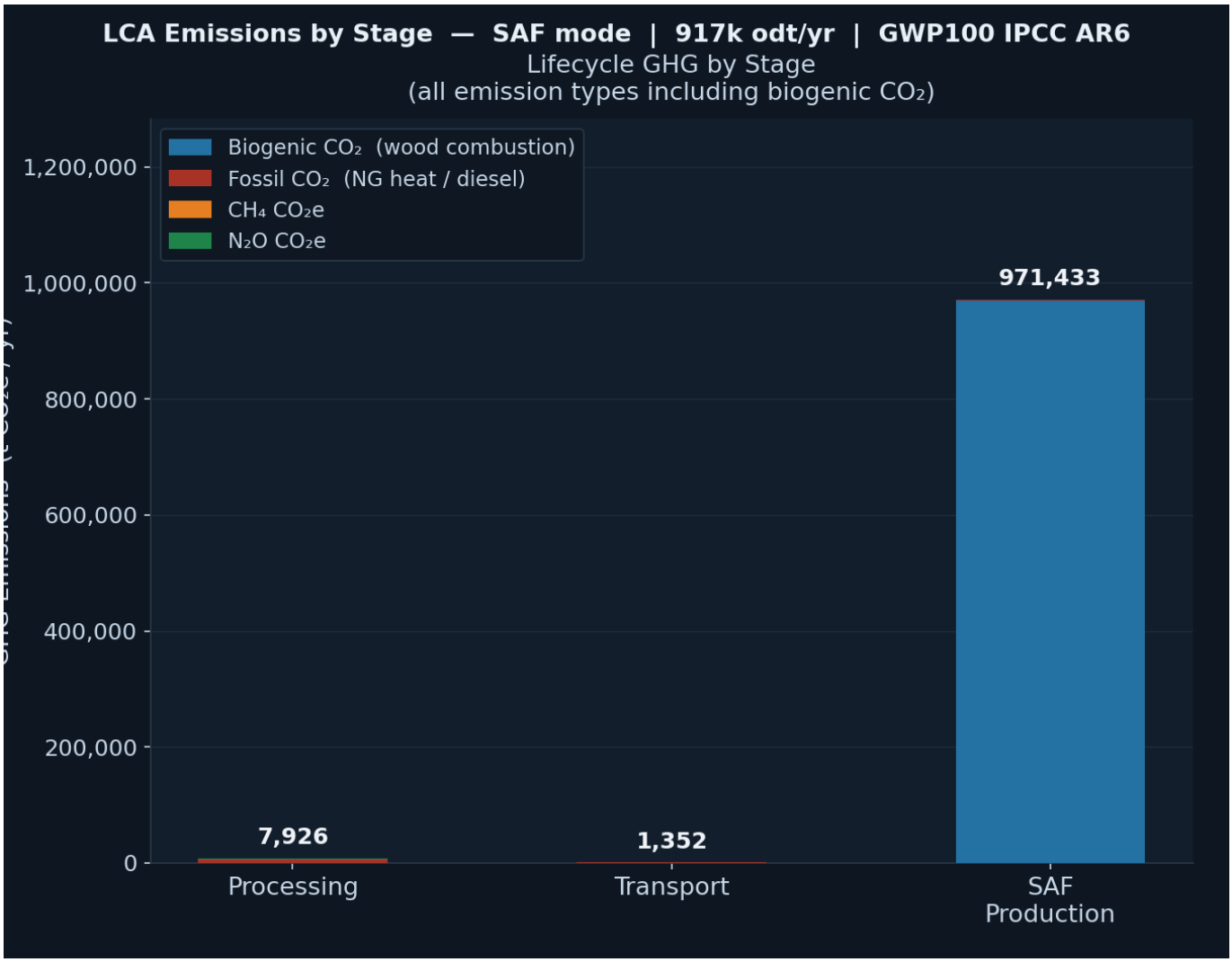


Figure 1. Lifecycle GHG emissions by stage.

4. Avoided Emissions Analysis

Baseline	Baseline GHG (t CO_2e/yr)	This Project (non-biogenic, t CO_2e/yr)	Avoided (t CO_2e/yr)	Reduction
Open Pile Burning	131,181	10,897	120,285	91.7%
Fossil Jet Fuel	58,542	10,897	47,645	81.4%

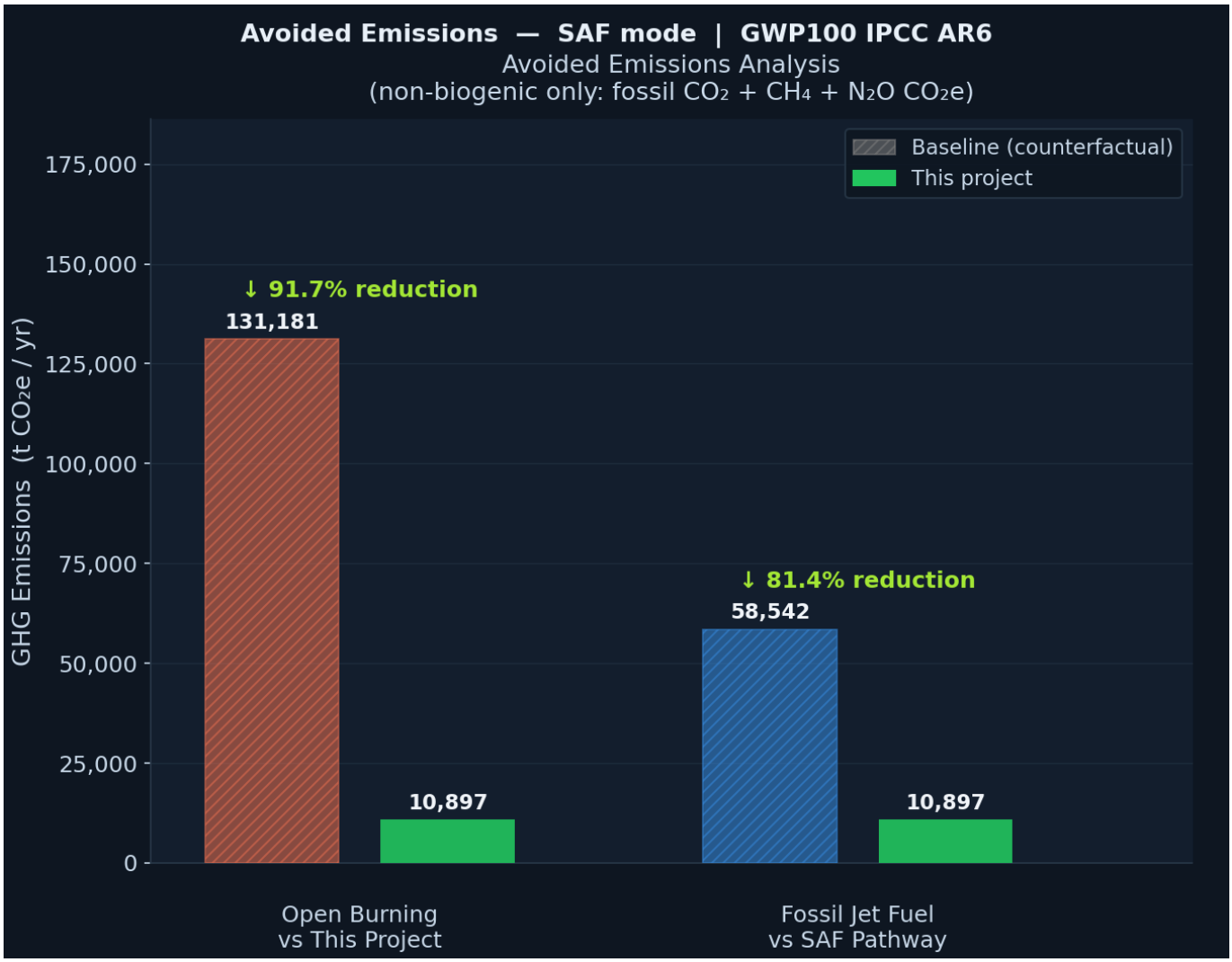


Figure 2. Avoided emissions — baseline (hatched) vs project (green). Non-biogenic only.

5. Policy Analysis

Policy Inputs (IRA 45Z)

Parameter	Value
SAF Sale Price	$6.0945/gal
Diesel Sale Price	$3.8990/gal
Naphtha Sale Price	$2.8391/gal
45Z SAF Credit	$0.350/GGE
45Z Non-SAF Credit	$0.200/GGE

Credit Duration	10 yr
Jet-A Market Price	$3.104/gal
Diesel Market Price	$5.072/gal
Naphtha Market Price	$3.710/gal
Solve Required Credit	No

Metric	No Policy	With 45Z Credit
NPV (Equity)	$-0.01M	$84.21M
Equity IRR	12.75%	15.81%
Payback	8.5 yr	6.5 yr
Revenue Yr1	$292.22M/yr	$292.22M/yr

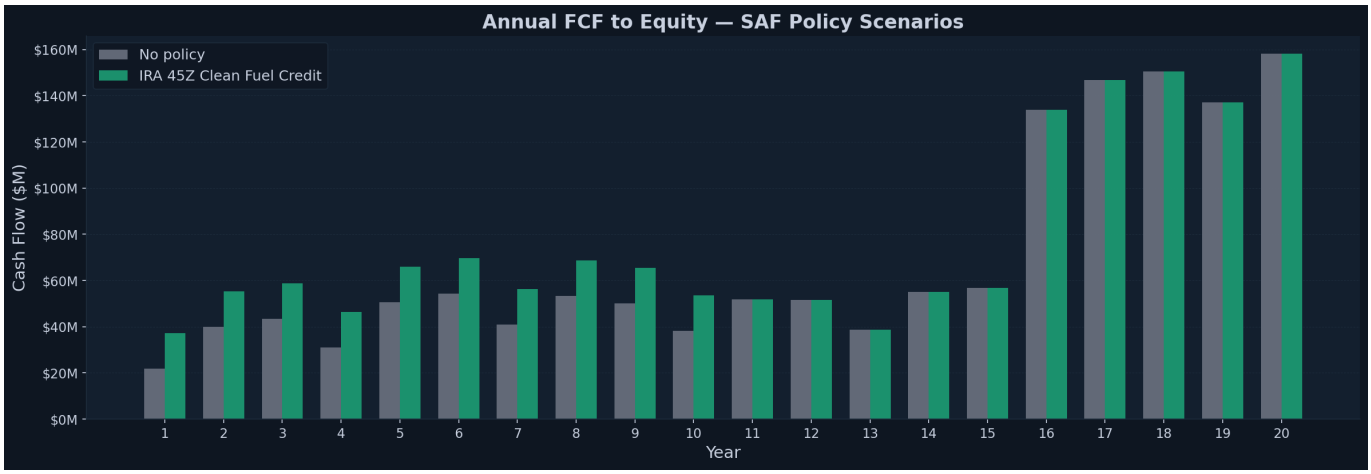


Figure 3. Policy scenario comparison.

6. Employment Impact

Category	Jobs	Description
Direct	47	Conversion facility operations
Indirect	436	Supply chain and services
Induced	233	Local economic spillover
TOTAL	**715**	

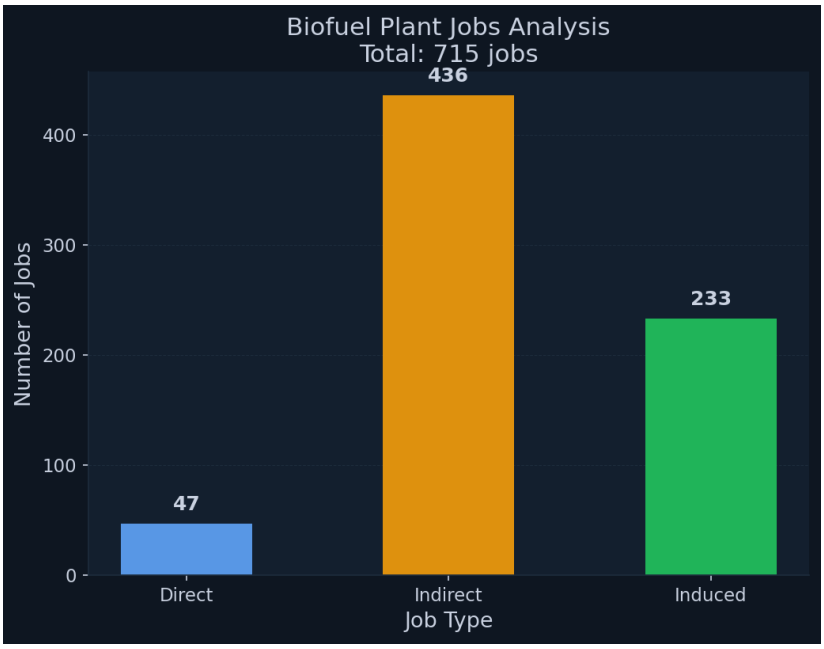


Figure 4. Employment impact by category (EPI sector multipliers).